

MBPS Pipeline: Verified Architecture & Adapter Survey

Corrected Methodology, Dimension Scaling Analysis, and PEFT Literature Review

Research Documentation

April 26, 2026

1 Critical Corrections to Previous Documentation

The following facts have been **verified by direct codebase inspection** of the MBPS project files in `/tmp/mbps_work/docs/`:

Claim	Previously (Incorrect)	Verified (Correct)
DCFA Backbone	DINOv3 ViT-B/16	DINOv2 ViT-B/14
Depth Model	Depth Anything v3	DepthPro
Training Data	COCO	Cityscapes (2,975 images)
Activation	GELU	ReLU
Output Init.	Residual skip	Zero-initialized + residual add
Total Params	40K or 147K	~225K (exact count below)
DCFA Architecture	106→384→90	106→384→384→90
SIMCF Features	Uses DINOv3 features	No DINO features used

Table 1: Corrected architectural facts from codebase verification.

2 Verified DCFA Architecture

2.1 Exact Parameter Count

DCFA v3 (V3_dd16_h384_l2):

Layer 1:	Linear(106 → 384)	= 106*384 + 384	= 41,088
LayerNorm 1:		= 384*2	= 768
Layer 2:	Linear(384 → 384)	= 384*384 + 384	= 147,840
LayerNorm 2:		= 384*2	= 768
Output:	Linear(384 → 90)	= 384*90 + 90	= 34,650

Total: 225,114 parameters

2.2 Training Configuration

- **Input:** 90-D CAUSE Segment_TR codes + 16-D sinusoidal depth encoding → 106-D
- **Hidden:** 384-D, 2-layer MLP
- **Activation:** ReLU (applied at each layer)
- **Output:** Linear(384→90), **zero-initialized** (starts as identity)
- **Skip:** $\hat{f} = f + \text{DCFA}(f, d)$
- **Loss:** $\mathcal{L} = \mathcal{L}_{\text{depth-corr}} + \lambda_{\text{preserve}} \|A_{\theta}(f, d) - f\|^2$ with $\lambda = 20.0$
- **Training data:** Cityscapes train set (2,975 images)

3 DCFA Dimension Scaling Analysis

The current DCFA outputs a 90-D vector because that is the CAUSE-TR code dimension. **Increasing the output dimension is architecturally possible** with the following tradeoffs:

3.1 Scaling Options

Hidden	Output	Total Params	Δ vs 90D	k-means	Assessment
384	90 (baseline)	225K	—	(80, 90)	Current
384	128	240K	+16K	(80, 128)	Minimal overhead
384	256	289K	+65K	(80, 256)	2x capacity, still small
384	512	388K	+164K	(80, 512)	Curse of dim. risk
512	128	385K	+160K	(80, 128)	Better hidden capacity
512	256	451K	+226K	(80, 256)	Dim. scaling concern
768	128	774K	+549K	(80, 128)	Large for adapter

Table 2: DCFA dimension scaling options with parameter counts.

3.2 Technical Recommendations

1. **Best first experiment:** 384 hidden $\rightarrow$ 256 output (+65K params). Doubles representational capacity without significant k-means degradation.
2. **With hidden expansion:** 512 hidden + 128 output (+160K). More nonlinear capacity, manageable output dimension.
3. **Avoid:** Output dim $>$ 512 with k=80 clusters. Curse of dimensionality degrades k-means separation.
4. **Alternative:** Keep 90D output but add *auxiliary depth feature projection* (16 $\rightarrow$ 32) concatenated before clustering, increasing effective cluster dimension without touching CAUSE codes.

4 Comprehensive Adapter Comparison for DCFA Replacement

4.1 Evaluation Criteria

For each adapter architecture from A* venues, we evaluate:

1. **Architecture fit:** Does it naturally replace a 106 $\rightarrow$ 90 cross-modal fusion MLP?
2. **Parameter budget:** Is it in the 200K–500K range or does it explode?
3. **Inference cost:** Mergeable (free) vs. adds latency?
4. **Proven on:** Classification vs. dense prediction vs. segmentation?

4.2 Complete Comparison Table

4.3 Verdict: Should DCFA Be Replaced?

No – DCFA is already well-designed for its specific role. Most surveyed adapters are designed for:

1. *Full ViT backbone fine-tuning* (LoRA, DoRA, AdaLoRA, AdaptFormer)
2. *Classification tasks* (VPT, FacT)
3. *Specific architectures* (SAM-Adapter, ViT-Adapter)

DCFA occupies a unique niche: **cross-modal feature fusion** (90D semantics + 16D depth) with **zero-initialized residual** output and **preservation loss**. No surveyed adapter directly addresses this specific design point.

5 Recommended Ablations

Based on verified architecture and literature survey:

Method	Type	Structure	Params	Suitability
DCFA (Current)	Cross-modal Bottleneck	MLP(106->384->384->90) + preserve	225K	Baseline. Depth-cond., zero-init stable.
DoRA (ICML 2024)	Weight-Decomposed LoRA	$W' = W + BA$, magnitude + direction	LoRA + d	A: Best LoRA variant. Magnitude/direction decomp.
Adapter+ (CVPR 2024)	Bottleneck + Scaling	Post-FFN adapter + learnable scaling	0.2M	A-: Best param-accuracy. Per-layer refactor.
LoRA (ICLR 2022)	Low-rank Reparam.	$W' = W + BA$, rank $r \ll d$	0.1–1M	B+: Good for transformers; limited for fusion.
AdaptFormer (NeurIPS 2022)	Parallel Adapter	$x' = x + s * \text{Adapter}(\text{LN}(x))$	0.5–1M/layer	B+: Proven for ViTs. Better for Stage-2 backbone.
Dynamic Tuning (NeurIPS 2024)	Dynamic Token + MoE	Token dispatcher + MoE adapter	0.5M + routing	B: 71% FLOP reduction. Overkill for DCFA.
AdaLoRA (ICLR 2023)	Adaptive Rank LoRA	$\Delta W = PAQ$, prune Λ	Budget-scheduled	B: Overkill for 225K adapter. Full-model budget.
LST (NeurIPS 2022)	Side Network + Gating	Side branch + learnable blending	1–2M	B-: Adds unnecessary complexity at DCFA scale.
VPT (ECCV 2022)	Visual Prompt Tuning	Prepend trainable prompt tokens	<1%	C+: Global prompts. Needs per-patch depth cond.
LoRA-FA	Fixed-A LoRA	Freeze A, train B; store XA activation	Half LoRA	C+: Memory optimization. Irrelevant at 225K scale.
ViT-Adapter (ICLR 2023)	Spatial Pyramid	Multi-scale conv $\rightarrow$ deform. attention	3–5M	C: Wrong granularity. Image-level, not per-patch.
FacT (AAAI 2023)	Tensorized Decomp.	3D tensor factors	8–50K	C: For full ViT. Not for standalone MLP.
SAM-Adapter (ICCV 2023)	Task-Specific Adapter	Multi-layer SAM encoder adapters	2–5M	C: SAM-specific. DCFA is fusion, not encoder.
ExPLoRA	Ext. Pre-train + LoRA	Unfreeze 1–2 blocks + LoRA rest	10–20M	C: Domain adaptation, not fusion.

Table 3: Adapter architectures evaluated for DCFA replacement (single table).

ID	Ablations to Run	Expected Impact	Effort
A1	Output dim: 90 $\rightarrow$ 128 $\rightarrow$ 256	+0.3–0.8 PQ	Low
A2	Hidden dim: 384 $\rightarrow$ 512 $\rightarrow$ 768	+0.2–0.5 PQ	Low
A3	Depth encoding: 16 $\rightarrow$ 32 (more freq bands)	+0.1–0.3 PQ	Low
A4	Add 3rd MLP layer (106 $\rightarrow$ 384 $\rightarrow$ 384 $\rightarrow$ 384 $\rightarrow$ 90)	+0.2 PQ / risk overfit	Low
A5	Replace ReLU with GELU/SiLU	Minimal; try SiLU first	Low
A6	Add patch-to-patch attention in DCFA (transformer block)	+0.5–1.0 PQ	Medium
A7	Vary preservation loss λ : 5, 10, 20, 40	Tune optimal balance	Low
A8	Depth conditioning ablation: depth-only vs. depth+edge	Verify depth is binding	Low
A9	Concatenate DINOv2 768D features instead of 90D codes	Major redesign; high risk	High
A10	DoRA on CAUSE-TR projection head (reparam. W)	Novel; test if rank helps	Medium

Table 4: Proposed ablations ordered by expected impact and implementation effort.

6 Key Papers Downloaded

All papers saved to `/mnt/agents/output/papers/`:

Filename	Method	Venue
adaptformer.pdf	AdaptFormer	NeurIPS 2022
adalora.pdf	AdaLoRA	ICLR 2023
adapter_plus.pdf	Adapter+	CVPR 2024
compacter.pdf	Compacter	ICML 2022
cmx.pdf	CMX	2023
dora.pdf	DoRA	ICML 2024 (Oral)
dynamic_tuning.pdf	Dynamic Tuning	NeurIPS 2024
efft.pdf	EFFT	2023
explorer.pdf	ExPLoRA	2024
fact.pdf	FacT	AAAI 2023
film.pdf	FiLM	NeurIPS 2017
lora_fa.pdf	LoRA-FA	2023
lst.pdf	Ladder Side-Tuning	NeurIPS 2022
med_sam_adapter.pdf	Medical SAM Adapter	ICCV 2023 / MedIA
sam2_adapter.pdf	SAM2-Adapter	2024
sam_adapter.pdf	SAM-Adapter	ICCV 2023
side_tuning.pdf	Side-Tuning	ICLR 2022
vit_adapter.pdf	ViT-Adapter	ICLR 2023
vpeft_survey.pdf	V-PEFT Survey	2024
vpt.pdf	VPT	ECCV 2022

Table 5: Downloaded literature for manual reading.